

Cisco ACI L3Out Configuration Examples

[Bias-Free Language](#)

Updated: April 27, 2020

Chapter: Example 1 - An OSPF L3Out With Two External Routers

Chapter Contents

Example 1 - An OSPF L3Out With Two External Routers

Example Topology

Prerequisites

Create L3Out Using the Create L3Out Wizard

Review - Create L3Out Using the Create L3Out Wizard

Configure Advertise the BD Subnet with a Route Map

Verify the Contract

Change the OSPF Interface Level Parameters (Optional)

This chapter includes the following sections:



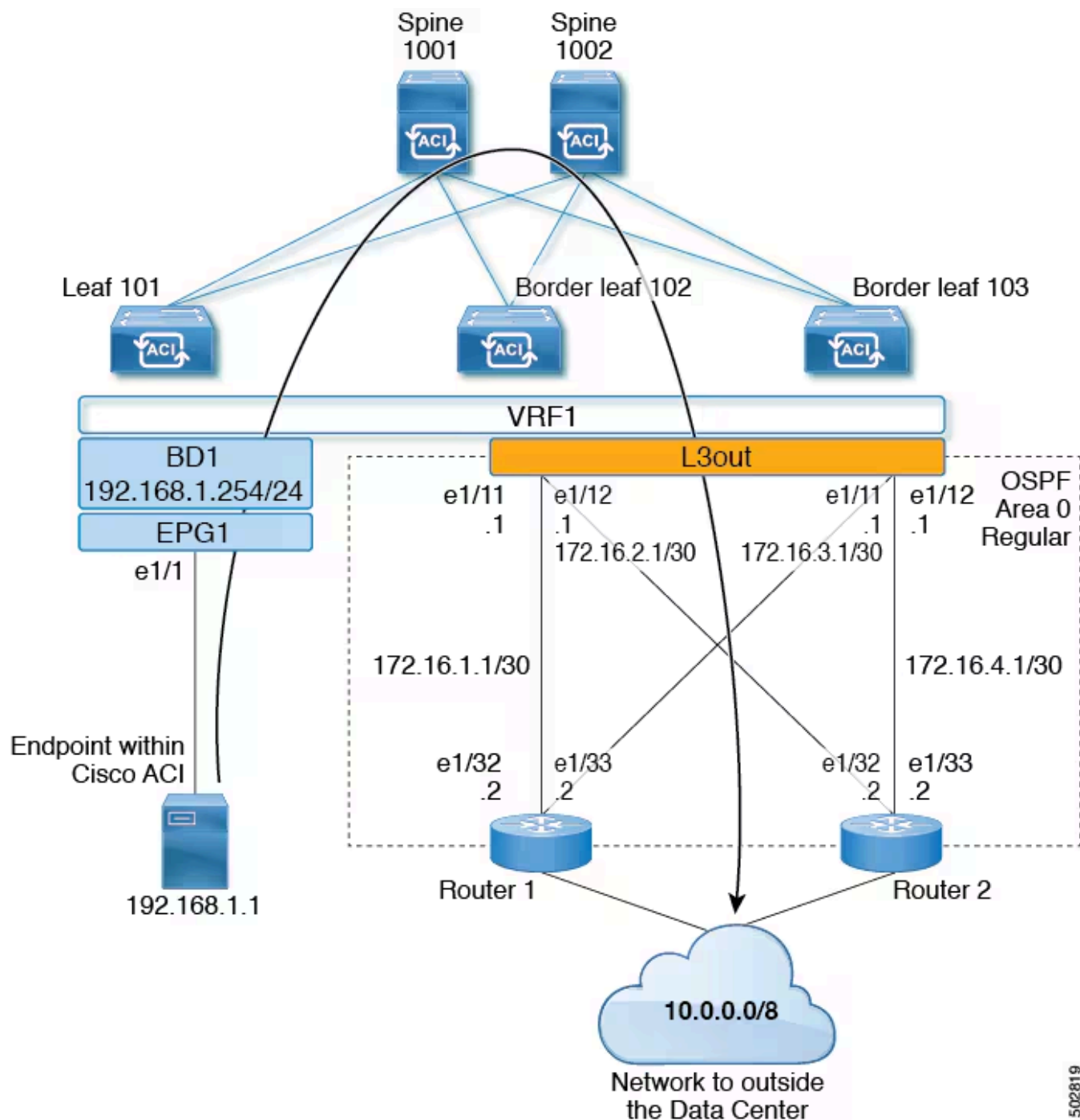
This example uses Cisco APIC release 4.2(x) and the associated GUI screens.

Note

- Example Topology
- Prerequisites
- Create L3Out Using the Create L3Out Wizard
- Review - Create L3Out Using the Create L3Out Wizard
- Configure Advertise the BD Subnet with a Route Map
- Verify the Contract
- Change the OSPF Interface Level Parameters (Optional)

Example Topology

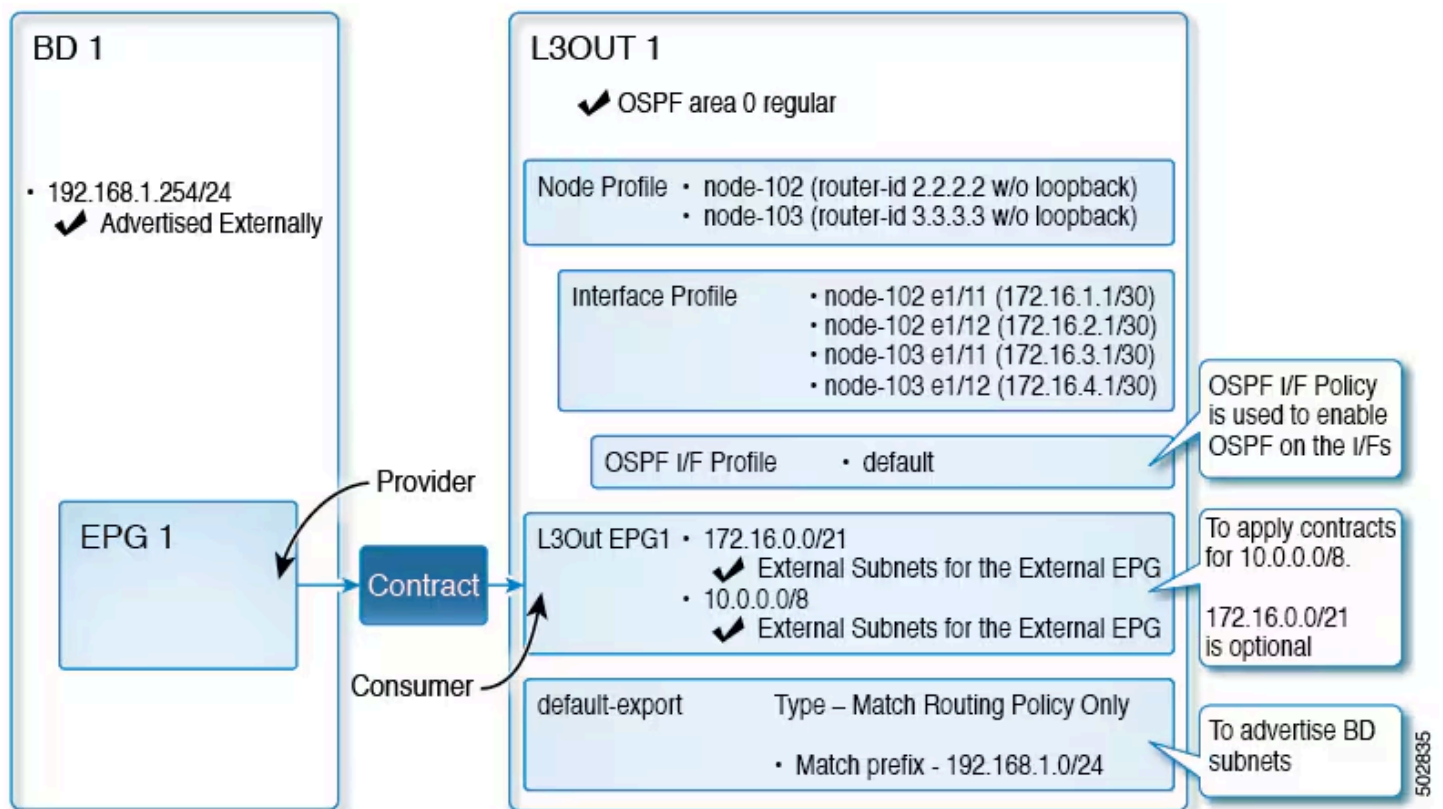
Figure 1. Example Topology for an OSPF L3Out with Two External Routers



This basic L3Out example shows you how to:

- Configure an L3Out with the following specifications
 - with Area 0 OSPF
 - with two external routers
 - with routed interfaces
 - on two border leaf switches
- Advertise a BD subnet using default route-map (default-export)
- Allow communication with a contract between EPG1 and external route (10.0.0.0/8)

Figure 2. OSPF Configuration Diagram



The preceding diagram illustrates the configuration for the example topology in Example Topology for an OSPF L3Out with Two External Routers. The configuration flow for this example is as follows:

1. L3Out: This creates
 - L3Out itself (OSPF parameters)
 - Node, Interface, OSPF I/F Profiles
 - L3Out EPG with **External Subnets for the External EPG** scope
2. Advertise a BD subnet: This uses
 - **default-export** route-map
 - BD subnet with **Advertise Externally** scope
3. Allow EPG - L3Out communication: This uses a contract between EPG1 and L3Out EPG1

Prerequisites

Figure 3. Example Screen of Objects Created as Prerequisites

Static Ports

Path	Primary VLAN for Micro-Seg	Port Encap (or Secondary VLAN for Micro-Seg)	Deployment Immediacy	Mode
Node: Pod-1				
Pod-1/Node-101...	unknown	vlan-10	Immediate	Trunk

IP Address: 192.168.1.254/24

Description: optional

Treat as virtual IP address: ☐

Make this IP address primary: ☐

Scope: ☒ Advertised Externally
☐ Shared between VRFs

Subnet Control: ☐ No Default SVI Gateway
☐ Querier IP

L3 Out for Route Profile: select a value

Route Profile: select a value

• This configuration example focuses only on the L3Out configuration part. The other configurations such as for VRF, BD, EPG, Application Profiles, and Access Policies (Layer 3 Domain etc.) are not covered. The preceding screenshot displays the prerequisite tenant configurations that are as follows:

- VRF1
- BD1 with the subnet 192.168.1.254/24
- EPG1 with a static port towards endpoints

See Completed Configurations in XML for REST API for a complete configuration example that includes VRF, BD, EPG, Application Profiles, and Access Policies on top of the L3Out configuration.

Create L3Out Using the **Create L3Out** Wizard

This task creates the OSPF L3Out described in Example Topology. Following this task, Cisco ACI will be configured with two border leaf switches and OSPF neighborhood with two external routers as shown in Example Topology for an OSPF L3Out with Two External Routers.

Procedure

- Step 1** In the GUI **Navigation** pane, under the Tenant Example, navigate to **Networking > L3Outs**.
- Step 2** Right-click and choose **Create L3Out**.

Step 3

In the **Create L3Out** screen, **Identity** tab, perform the following actions:

Create L3Out

1. Identity 2. Nodes And Interfaces 3. Protocols 4. External EPG

Leaf L Router R

Identity

A Layer 3 Outside (L3Out) network configuration defines how the ACI fabric connects to external layer 3 networks. The L3Out supports connecting to external networks using static routing and dynamic routing protocols (BGP, OSPF, and EIGRP).

Prerequisites:

- Configure an L3 Domain and Fabric Access Policies for interfaces used in the L3Out (AAEP, VLAN pool, Interface selectors).
- Configure a BGP Route Reflector Policy for the fabric infra MP-BGP.

Name: EXAMPLE_L3Out1

VRF: VRF1

L3 Domain: EXAMPLE_L3_DOM

Use for GOLP: ☐

☐ BGP ☐ EIGRP ☒ OSPF

OSPF Area ID: 0

OSPF Area Control: ☒ Send redistributed LSAs into NSSA area ☒ Originate summary LSA ☐ Suppress forwarding address in translated LSA

OSPF Area Type: NSSA area Regular area Stub area

OSPF Area Cost: 1

Previous Cancel Next

- In the **Name** field, enter the name for an L3Out. (EXAMPLE_L3Out1)
- In the **VRF** field and the **L3 Domain** field, choose the appropriate values. (VRF1, EXAMPLE_L3DOM)
- In the **OSPF** field, check the checkbox.
- In the **OSPF Area ID** field, choose the value **0** or the text **backbone**.
- In the **OSPF Area Type** field, choose **Regular area**.
- Keep the rest of the fields with their default values.

Step 4

Click **Next** to display the **Nodes and Interfaces** screen, and perform the following actions:

Create L3Out

Nodes and Interfaces

The L3Out configuration consists of node profiles and interface profiles. An L3Out can span across multiple nodes in the fabric. All nodes used by the L3Out can be included in a single node profile and is required for nodes that are part of a VPC pair. Interface profiles can include multiple interfaces. When configuring dual stack interfaces a separate interface profile is required for the IPv4 and IPv6 configuration, that is automatically taken care of by this wizard.

Use Defaults: ☒

Interface Types

Layer 3: **Routed** Routed Sub SVI Floating SVI

Layer 2: **Port** Direct Port Channel

Nodes

Node ID: leaf2 (Node-102) Router ID: 2.2.2.2 Loopback Address: **Hide Interfaces**

Interface: eth1/11 IP Address: 172.16.1.1/30 MTU (bytes): inherit **+**

Interface: eth1/12 IP Address: 172.16.2.1/30 MTU (bytes): inherit **+**

Node ID: leaf3 (Node-103) Router ID: 3.3.3.3 Loopback Address: **Hide Interfaces**

Interface: eth1/11 IP Address: 172.16.3.1/30 MTU (bytes): inherit **+**

Interface: eth1/12 IP Address: 172.16.4.1/30 MTU (bytes): inherit **+**

Previous Cancel Next

- In the **Interface Types** area, in the **Layer 3** field and in the **Layer 2** field, ensure that your selections match the choices in the preceding screenshot (Routed and Port).
- In the **Nodes** area, in the **Node ID** field, from the drop-down list, choose the appropriate node ID. (leaf2 (Node 102))
- In the **Router ID** field, enter the appropriate router ID. (2.2.2.2)
The **Loopback Address** field auto populates based on the router ID value you enter. You do not require the loopback address, so delete the value and leave the field blank.
- In the **Interface** field, choose the interface ID. (eth1/11)
- In the **IP Address** field, enter the associated IP address. (172.16.1.1/30)
- In the **MTU** field, keep the default value. (inherit)
- Click the **+** icon next to the **MTU** field to add an additional interface for node leaf2. (Node-102)
- In the **Interface** field, choose the interface ID. (eth1/12)
- In the **IP Address** field, enter the associated IP address. (172.16.2.1/30)
- In the **MTU** field, keep the default value. (inherit)

Step 5

To add another node, click the **+** icon next to the **Loopback Address** field, and perform the following actions:

Note When you click the **+** icon, the new **Nodes** area is displayed below the area that you had populated earlier.

- a. In the **Nodes** area, in the **Node ID** field, from the drop-down list, choose the node ID. (leaf3 (Node-103))
- b. In the **Router ID** field, enter the router ID. (3.3.3.3)
The **Loopback Address** field auto populates based on the router ID value you enter. You do not require the loopback address, so delete the value and leave the field blank.
- c. In the **Interface** field, choose the interface ID. (eth1/11)
- d. In the **IP Address** field, enter the IP address. (172.16.3.1/30)
- e. In the **MTU** field, keep the default value. (inherit)
- f. Click the **+** icon next to the **MTU** field to add an additional interface for node leaf3. (Node-103)
- g. In the **Interface** field, choose the interface ID. (eth1/12)
- h. In the **IP Address** field, enter the associated IP address. (172.16.4.1/30)
- i. In the **MTU** field, keep the default value. (inherit), and click **Next**.

We have specified the node, interface, and IP address for each interface.

Step 6

Click **Next** to view the **Protocols** screen.

This screen allows you to specify the OSPF interface level policy to configure hello-interval, network-type, etc.

Create L3Out

Protocol Associations

OSPF

Node ID	Interface	Policy	Hide Policy
Node ID: 102	1/11,1/12	select a value	☐
Node ID: 103	1/11,1/12	select a value	☐

Previous Cancel Next

In this example, nothing is selected. Therefore, the default policy is used. The default OSPF interface profile uses **Unspecified** as network-type which defaults to broadcast network type. To optimize this with point-to-point network-type for sub-interface, see **Change the OSPF Interface Level Parameters (Optional)**.

Step 7

Click **Next**.

The **External EPG** screen is displayed with L3Out EPG details. This configuration is to classify the traffic into the EPG to apply to the contract.

Step 8 In the **External EPG** screen, perform the following actions:

Create L3Out

External EPG

The L3Out Network or External EPG is used for traffic classification, contract associations, and route control policies. Classification is matching external networks to this EPG for applying contracts. Route control policies are used for filtering dynamic routes exchanged between the ACI fabric and external devices, and leaked into other VRFs in the fabric.

Name: L3Out_EPG1

Provided Contract: select a value

Consumed Contract: common/default

Default EPG for all external networks: ☐

Subnets

IP Address	Scope	Name	Aggregate	Route Control Profile	Route Summarization Policy
10.0.0.0/8	External Subnets for the External EPG				
172.16.0.0/21	External Subnets for the External EPG				

Previous Cancel Finish

- In the **External EPG** area, **Name** field, enter a name for the external EPG. (L3Out_EPG1)
- In the **Provided Contract** field, do not choose a value.
In this example, there is no provided contract for L3Out_EPG1 because a normal EPG (EPG1) is the provider.
- In the **Consumed Contract** field, choose **default** from the drop-down list.

Step 9 In the **Default EPG for all external networks** field, uncheck the checkbox, and perform the following actions:

- Click the **+** icon in the **Subnets** area, to display the **Create Subnet** dialog box.
- In the **IP Address** field, enter the subnet. (10.0.0.0/8)
- In the **External EPG Classification** field, check the checkbox for **External Subnets for the External EPG**. Click **OK**.

Step 10 Click the **+** icon in the **Subnets** area once more to display the **Create Subnet** dialog box, and perform the following actions:

Note Although this is an optional configuration, it is a best practice to specify the L3Out interface subnets in case endpoints have to communicate with those IPs.

- In the **IP Address** field, enter the subnet. (172.16.0.0/21)
This subnet covers all the interfaces in the L3Out. This can be each individual subnet for each routed interface instead.
- In the **External EPG Classification** field, check the checkbox for **External Subnets for the External EPG**. Click **OK**.
- Click **Finish**.

The L3Out OSPF is now deployed.

Review – Create L3Out Using the **Create L3Out** Wizard

Review how the configuration using the wizard is presented in the Cisco APIC GUI, and verify that the configurations are accurate.

Procedure

- Step 1** Navigate to your **Tenant_name** > **Networking** > **L3Outs** > **EXAMPLE_L3Out1**, in the **Work** pane, scroll to view the details as follows:
- At this location in the GUI, verify the main L3Out parameters such as VRF, domain, and OSPF parameters that are configured in the **Identity** screen in the **Create L3Out** wizard.

The screenshot displays the configuration of **EXAMPLE_L3Out1** in the **Work** pane. The left pane shows the navigation tree with **EXAMPLE_L3Out1** selected. The right pane shows the configuration details for VRF, L3 Domain, OSPF parameters, Nodes, and Routed Interfaces.

VRF: VRF1
Resolved VRF: Example/VRF1
L3 Domain: EXAMPLE_L3_DOM
Route Profile for Interleak: select a value

Enable BGP/EIGRP/OSPF: ☐ BGP ☒ OSPF ☐ EIGRP
OSPF Area ID: 0
OSPF Area Control: ☒ Send redistributed LSAs into NSSA area ☒ Originate summary LSA ☐ Suppress forwarding address in translated LSA
OSPF Area Type: NSSA area Regular area Stub area
OSPF Area Cost: 1

Nodes:

Node ID	Router ID	Loopback Address
topology/pod-1/node-102	2.2.2.2	
topology/pod-1/node-103	3.3.3.3	

Routed Interfaces:

Path	IP Address	Secondary IP Address	MAC Address	MTU (bytes)
Pod-1/Node-102/eth1/11	172.16.1.1/30		00:22:BD:F8:19:FF	Inherit
Pod-1/Node-102/eth1/12	172.16.2.1/30		00:22:BD:F8:19:FF	Inherit
Pod-1/Node-103/eth1/11	172.16.3.1/30		00:22:BD:F8:19:FF	Inherit
Pod-1/Node-103/eth1/12	172.16.4.1/30		00:22:BD:F8:19:FF	Inherit

- Step 2** Verify that OSPF is enabled with the specified parameters such as Area ID and Area Type.
- Step 3** Under **Logical Node Profiles**, *EXAMPLE_L3Out1_nodeProfile* is created to specify border leaf switches with their router IDs.
- Step 4** Under **Logical Interface Profile**, *EXAMPLE_L3Out1_interfaceProfile* is created.
- Verify the interface parameters such as interface ID, IP addresses, in this example, as routed interfaces. The default MAC addresses gets auto populated. OSPF interface profile is also created under this for OSPF interface level parameters.

The review is complete.

Configure Advertise the BD Subnet with a Route Map

In this example, a route map, **default-export**, is used with the IP prefix list to advertise the BD subnet.

- Note** This **default-export** route map will be applied to the L3Out (EXAMPLE_L3Out1) without being associated to anything specific.

Procedure

Step 1 To enable a BD subnet to be advertised, navigate to **Tenant > Networks > Bridge Domains > BD1 > Subnets > 192.168.1.254/24**, and select **Advertised Externally** scope.

Example

Bridge Domains

BD1

DHCP Relay Labels

Subnets

192.168.1.254/24

ND Proxy Subnets

VRFs

External Bridged Networks

L3Outs

EXAMPLE_L3Out1

Logical Node Profiles

EXAMPLE_L3Out1_nodeProfile

Logical Interface Profiles

EXAMPLE_L3Out1_interfacePr...

OSPF Interface Profile

Configured Nodes

External EPGs

Route map for import and export route c...

default-export

Dot1Q Tunnels

Contracts

Policies

Services

Subnet - 192.168.1.254/24

PolicyOperationalFaultsHistory

Properties

IP Address: 192.168.1.254/24

Description: optional

Treat as virtual IP address: ☐

Make this IP address primary: ☐

Scope: ☐ Private to VRF ☒ Advertised Externally ☐ Shared between VRFs

Subnet Control: ☐ No Default SVI Gateway ☐ Querier IP

L3 Out for Route Profile: select a value

Route Profile: select a value

Show UsageResetSubmit

Step 2 To create a route map under your L3Out (EXAMPLE_L3Out1), navigate to **Route map for import and export route control**.

Example

Quick Start

Example

Application Profiles

Networking

Bridge Domains

VRFs

External Bridged Networks

L3Outs

EXAMPLE_L3Out1

Logical Node Profiles

EXAMPLE_L3Out1_nodeProfile

Logical Interface Profiles

EXAMPLE_L3Out1_interfacePr...

OSPF Interface Profile

Configured Nodes

External EPGs

L3Out_EPG1

Route map for import and export route c...

default-export

Dot1Q Tunnels

Contracts

Policies

Services

Name: default-export

Type: Match Prefix AND Routing Policy

Description: optional

Contexts

Order	Name	Action	Description
0	BD_Subnets	Permit	

Order: 0

Name: BD_Subnets

Action: Deny Permit

Description: optional

Match Rule: BD1_prefix

Set Rule: select a value

Match Prefix:

IP	Description	Aggregate
192.168.1.0/24		False

- Step 3** Right-click and choose **Create Route map for import and export route control**.
- Step 4** In the **Create Route map for import and export route control** dialog box, in the **Name** field, choose **default-export**.
- Step 5** In the **Type** field, choose **Matching Route Policy Only**.
- Note** **Match Routing Policy Only:** By choosing this **Type** with default-export route map, all route advertisement configuration is performed by this route map. BD associations and export route control subnets configured under the external EPG will not apply. You should configure all match rules within this route-map for all routes that will be advertised from this L3Out.
- Match Prefix and Routing Policy:** By choosing this **Type** with default-export route map, route advertisement is matched by any match rules configured in this route map **in addition to** any BD to L3Out associations and export route control subnets defined under the External EPG.
- When using a route profile, it is recommended to use **Match Routing Policy Only** for a simpler configuration that is easier to maintain.
- Step 6** In the **Contexts** area, click the **+** icon, to display the **Create Route Control Context** dialog box, and perform the following actions:
- In the **Order** field, configure the order. (0)
In this example, we have only one order.
 - In the **Name** field, enter a name for the context. (BD_Subnets)
 - In the **Action** field, choose **Permit**.
This enables the route map to permit the prefix we will configure.
- In this example, we require the match rule that requires the IP prefix list, **BD1_prefix**. This IP prefix list points to the BD subnet advertised.
- Step 7** In the **Match Rule** field, create the IP prefix-list by performing the following actions;
- Choose **Create Match Rule for a Route-Map**.
 - In the **Name** field, enter a name *BD1_prefix*.
 - In the **Match Prefix** area, click the **+** icon, and enter the BD subnet (192.168.1.0/24).

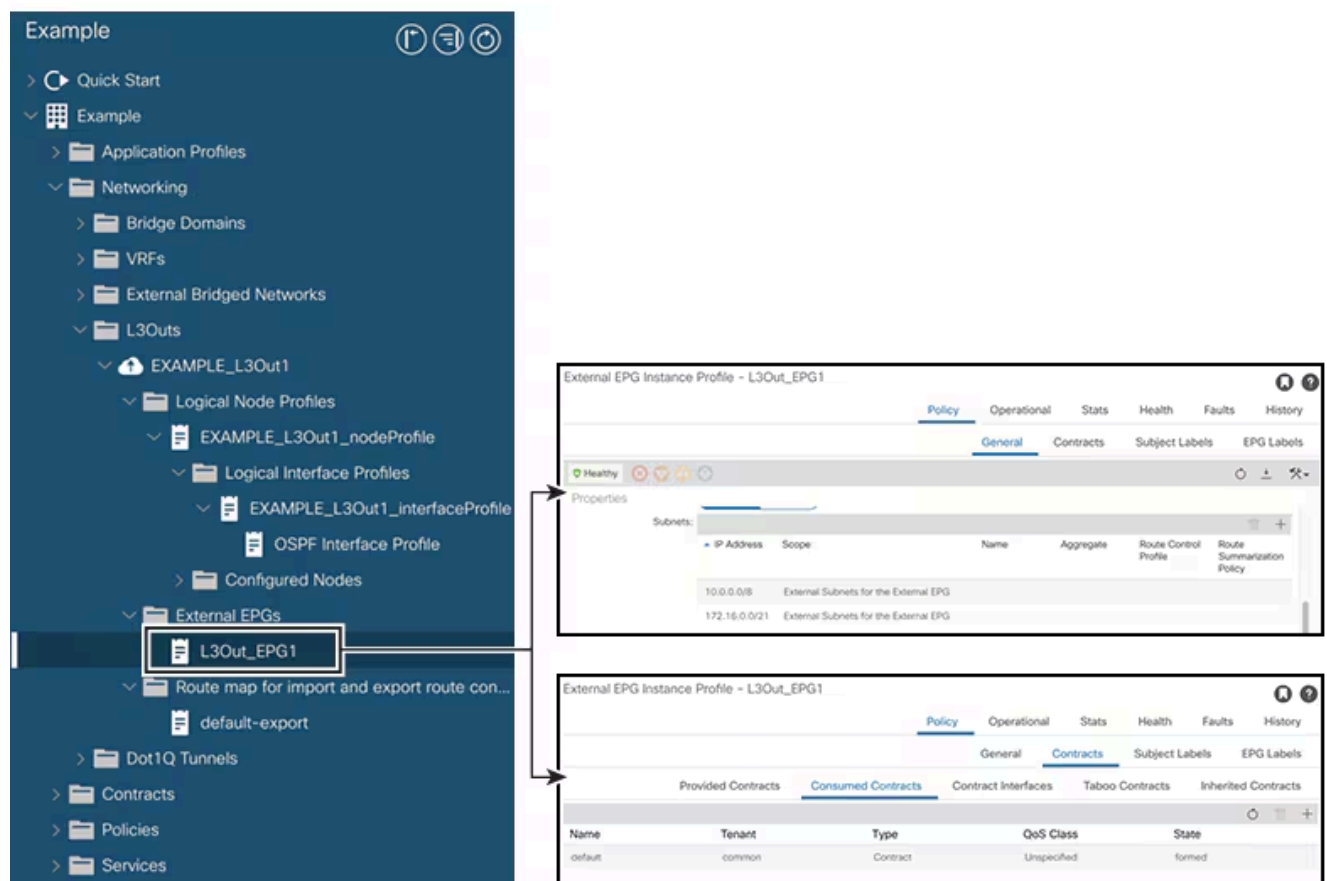
Verify the Contract

In this task, you verify the contract to enable communication between an endpoint (192.168.1.1) and external prefixes (10.0.0.0/8, and optionally 172.16.0.0/21). In this example, the EPG for the endpoint is EPG1 and the external EPG for external prefixes is L3Out_EPG1.

The required configuration should already be present from the **Create L3Out** wizard.

Procedure

- Step 1** Under your L3Out, navigate to **External EPGs > L3Out_EPG1**.



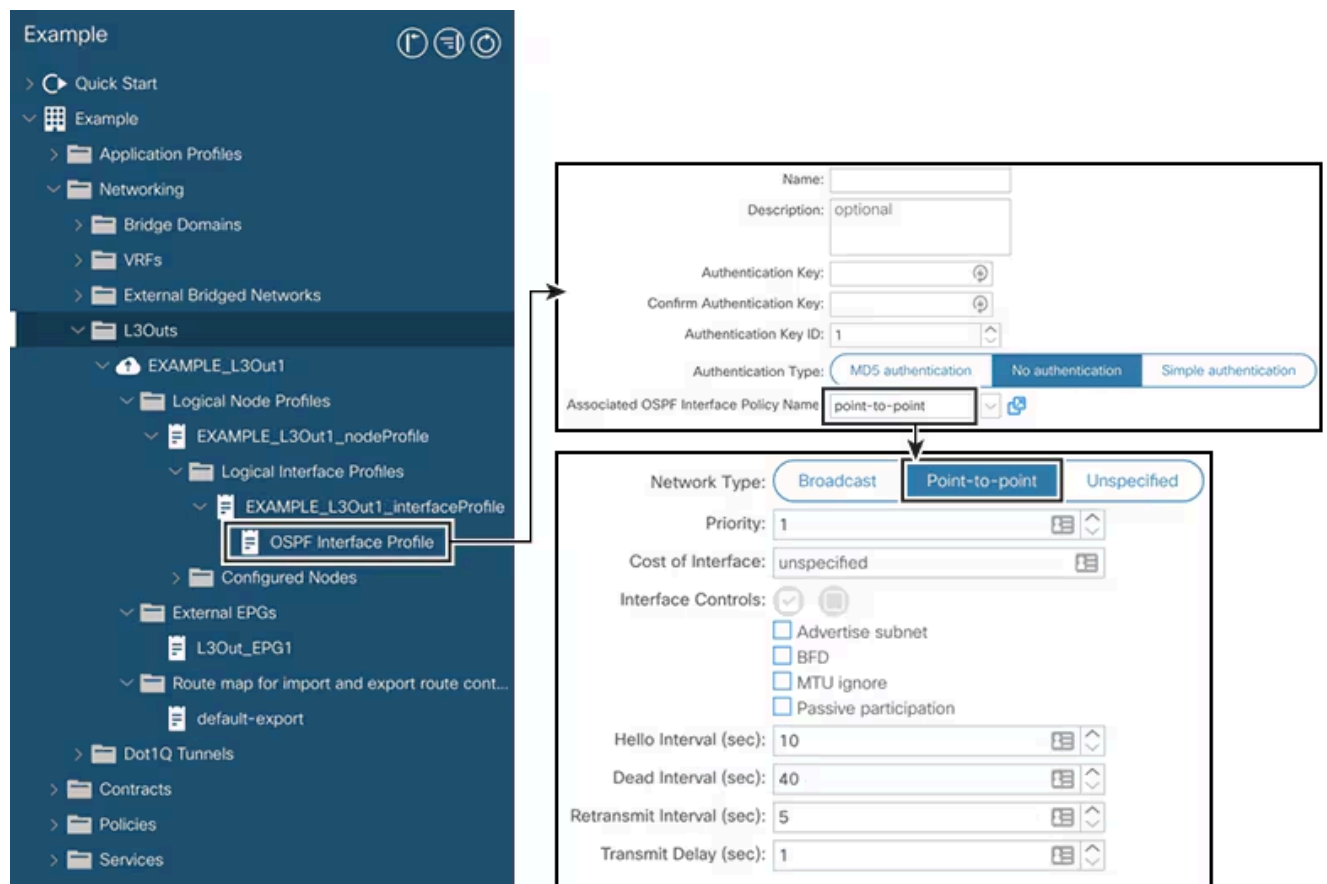
- Step 2** In the **Work** pane, in the **External EPG Instance Profile** area, under **Policy** > **General** sub-tab, look at the Properties and verify that the two subnets are displayed with **External Subnets for the External EPG**.
- Step 3** Next, click the **Contracts** sub-tab and verify the contract you specified earlier is consumed correctly. In case you want to add more contracts, you can perform the actions from this location in GUI.
- Step 4** Navigate to **Application Profile** > **Application EPGs** > **EPG1** > **Contracts**, and verify that EPG1 is providing the appropriate contract.

Change the OSPF Interface Level Parameters (Optional)

If you wish to change the OSPF interface-level parameters, such as **Hello Interval**, **OSPF network type**, then you can configure it in the OSPF Interface Profile. The node level OSPF parameters are already configured.

Procedure

- Step 1** Under your L3Out, navigate to **Logical Interface Profile** > **EXAMPLE_L3Out1_interfaceProfile** > **OSPF Interface Profile**.



Step 2 In the **Work** pane, in the **Properties** area, choose the OSPF Interface Policy you wish to use.

This modifies your OSPF interface level parameters.

Quick Links

[About Cisco](#)

[Contact Us](#)

[Careers](#)

[Connect with a partner](#)

Resources and Legal

[Feedback](#)

[Help](#)

[Terms & Conditions](#)

[Privacy](#)

[Cookies / Do not sell or share my personal data](#)

[Accessibility](#)

[Trademarks](#)

[Supply Chain Transparency](#)

[Newsroom](#)

[Sitemap](#)



© 2026 Cisco Systems, Inc.
